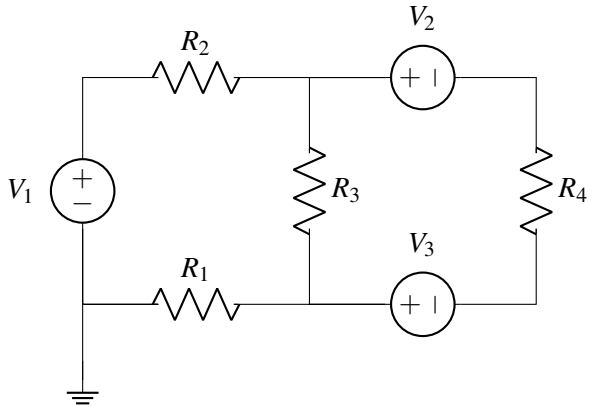
CSM
Fall 2026

EE 64
Worksheet 3

1. Supernode

Solution: Prereq: Best placed after a super easy nodal analysis problem.
Description: This shows how to deal with multiple voltage sources that don't share a ground when doing nodal analysis since we don't know how much current goes through a voltage sources.

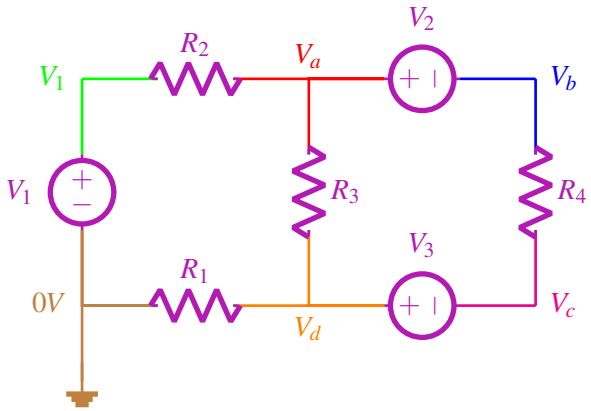
In this question, we will explore how to deal with multiple voltage sources when doing nodal analysis. Throughout the problem, assume that $R_1 = 1\Omega, R_2 = 2\Omega, R_3 = 4\Omega, R_4 = 4\Omega, V_1 = 1\text{V}, V_2 = 2\text{V}, V_3 = 4\text{V}$. While it is good practice to symbolically solve problems before substituting specific numbers, for superposition it is often easiest to substitute before superposing the solwers.



(a) Mark all the nodes. If you know the potential at the node, write down the value next to the node. If you don't know the value, then assign a variable for the potential.

Solution:

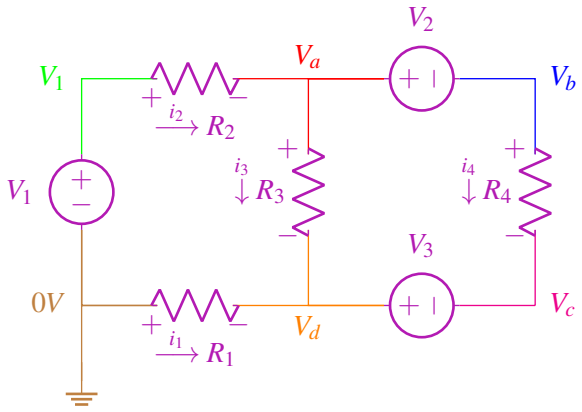
V_a, V_b, V_c, V_d are variables. We know the potentials at the node marked ground and at the node marked V_1.



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(b) Mark current directions arbitrarily and corresponding polarities on each resistor. Note that if current goes from left to right, then the left side of the resistor is to be marked + and the right side must be marked -. This is the passive sign convention.

Solution:



Solution: Might be a good idea to mark the nodes in the same fashion as the solutions for later parts!

(c) Note that we define 4 nodes with unknown potentials. So we need 4 equations. Each of these nodes with unknown potential should give us one equation.
Write the equation for the first node with unknown potential.

Solution: For the V_a node, we have a problem. The problem is that we don't know what current enters the voltage source V_2. The fix here is to treat V_a and V_b combined as a super-node. Since we are considering 2 nodes at the same time, this 'supernode' better give us 2 equations!
We can get equation from the fact that the voltage source exists.

$$V_a - V_b = V_2 \tag{1}$$

For the other equation, let's consider: (Sum of current entering nodes V_a and V_b) = (Sum of current leading nodes V_a and V_b).

Consider V_a:

- i_2 enters V_a.
- i_3 leaves V_a.

Consider V_b:

- i_4 leaves V_b.

$$i_2 = i_3 + i_4$$

$$\frac{V_1 - V_a}{R_2} = \frac{V_a - V_d}{R_3} + \frac{V_b - V_c}{R_4} \tag{2}$$

Solution: For equation 2, make sure students know that in the numerator, you have to write it as $V_+ - V_-$, depending on how you marked the polarities. But it is super important to mark the + where the current starts and - where it ends. This is passive sign convention. Make sure they do this right! Also, if students ask why (Sum of current entering nodes V_a and V_b) = (Sum of current leading nodes V_a and V_b) is something that we can use, show on the circuit that current cannot stay between V_a and V_b. Any current that enters V_a or V_b must leave through V_a or V_b

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(d) Write equations for all remaining unknown nodes. Solve for all the unknown potentials.

Solution: Nodes V_c and V_d remain. Let's start at V_c. Again, we don't know what current enters V_3, so we do the same thing. Treat the 2 ends of the voltage source, V_c and V_d as a super-node. One equation we get is

$$V_d - V_c = V_3 \tag{3}$$

The other equation comes from (Sum of current entering nodes V_a and V_b) = (Sum of current leading nodes V_a and V_b).

Consider V_c:

- i_4 enters.

Consider V_d:

- i_3 enters
- i_1 enters.

$$i_4 + i_3 + i_1 = 0$$
$$\frac{V_b - V_c}{R_4} + \frac{V_a - V_d}{R_3} + \frac{0 - V_d}{R_1} = 0 \tag{4}$$

Equations (1), (2), (3), (4) are four equations for the four variables V_a, V_b, V_c, V_d and can be solved uniquely. One fun fact is that these equations are guaranteed to be linearly independent as long as the circuit is not an impossible circuit (such as 2 current sources of differing values in series, or 2 voltage sources of different values in parallel).

To summarize, we have:

$$\begin{cases} V_a - V_b = V_2 \\ \dfrac{V_1 - V_a}{R_2} = \dfrac{V_a - V_d}{R_3} + \dfrac{V_b - V_c}{R_4} \\ V_d - V_c = V_3 \\ \dfrac{V_b - V_c}{R_4} + \dfrac{V_a - V_d}{R_3} + \dfrac{0 - V_d}{R_1} = 0 \end{cases} \Longrightarrow \begin{cases} V_a - V_b = 2 \\ \dfrac{1 - V_a}{2} = \dfrac{V_a - V_d}{4} + \dfrac{V_b - V_c}{4} \\ V_d - V_c = 4 \\ \dfrac{V_b - V_c}{4} + \dfrac{V_a - V_d}{4} + \dfrac{-V_d}{1} \end{cases}$$

Solving the right set of equations, we find that $\boxed{V_a = \frac{1}{5}\text{V}, V_b = -\frac{9}{5}\text{V}, V_c = -\frac{18}{5}\text{V}, V_d = \frac{2}{5}\text{V}}$.

(e) Now we will explore solving the circuit using superposition. To recall the process, which 2 sources will be suppressed when looking at each of the 3 sources. Will the other 2 sources be treated as open circuits or short circuits?

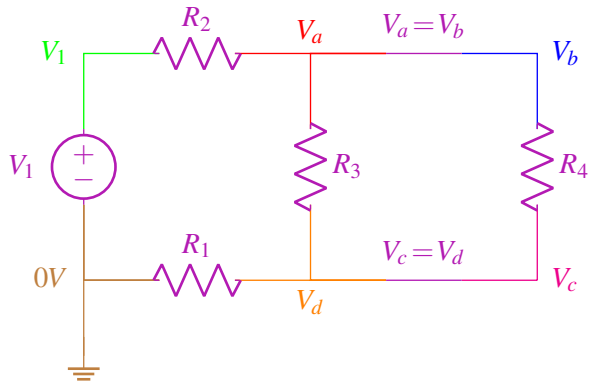
Solution: For V_1, V_2 and V_3 will be shorted. For V_2, V_1 and V_3 will be shorted. For V_3, V_1 and V_2 will be shorted. When doing superposition and suppressing inputs, a voltage source is treated as a short and a current source is treated as an open.

The reasoning for "zeroing" voltage/current sources is as follows: we define a voltage source as something that maintains a fixed voltage drop between the two terminals. If we "zero" that source, we are saying the voltage across the two terminals should be 0V. But that, by definition, has the same effect as connecting the two terminals with a wire, i.e. shorting it! So we say "zeroed" voltage sources are shorted. A similar argument leads us to find "zeroed" current sources are equivalent to open circuits.

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(f) Draw the superposition circuit for V_1 and solve this circuit.

Solution:



$$\begin{cases} \dfrac{V_a - V_1}{R_2} + \dfrac{V_a - V_d}{R_3} + \dfrac{V_b - V_c}{R_4} = 0 \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_c - V_b}{R_4} = 0 \\ V_a = V_b \\ V_c = V_d \end{cases} \qquad \Longrightarrow \qquad \begin{cases} \dfrac{V_a - V_1}{R_2} + \dfrac{V_a - V_d}{R_3} + \dfrac{V_a - V_d}{R_4} = 0 \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_d - V_a}{R_4} = 0 \end{cases}$$

Solving for $V_a, V_b, V_c,$ and V_d we find

$$\begin{cases} V_a = V_b = \dfrac{(R_1R_2 + R_1R_4 + R_3R_4)}{R} V_1 \\ V_d = V_c = \dfrac{R_1R_3 + R_1R_4}{R} V_1, \end{cases} \qquad R \equiv R_1R_3 + R_2R_3 + R_1R_4 + R_2R_4 + R_3R_4$$

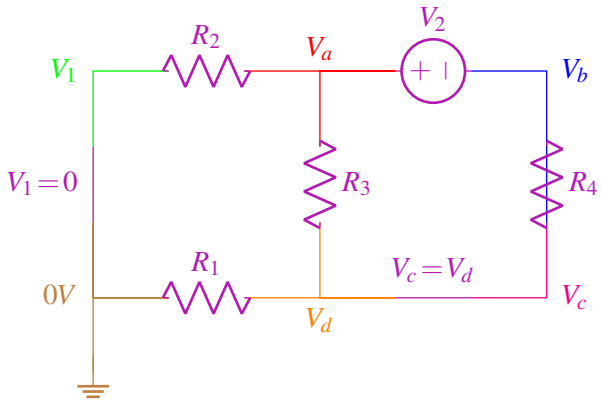
Or, if you prefer solving with numbers first:

$$\begin{cases} \dfrac{V_a - 1}{2} + \dfrac{V_a - V_d}{4} + \dfrac{V_a - V_d}{4} = 0 \\ \dfrac{V_d}{1} + \dfrac{V_d - V_a}{4} + \dfrac{V_d - V_a}{4} = 0 \end{cases} \Longrightarrow \boxed{\begin{cases} V_a = \frac{3}{5} \\ V_b = \frac{3}{5} \\ V_c = \frac{1}{5} \\ V_d = \frac{1}{5} \end{cases}}$$

(g) Draw the superposition circuit for V_2 and write the equation that characterizes this circuit.

Solution:

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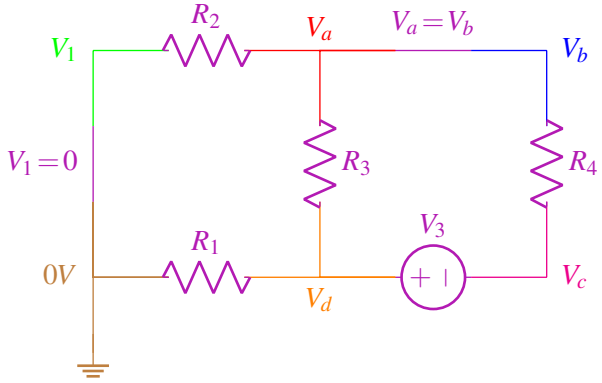


$$\begin{cases} \dfrac{V_1 - V_a}{R_2} + \dfrac{V_1 - V_d}{R_1} = 0 \implies V_a = -2V_d \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_c - V_b}{R_4} = 0 \implies 5V_d - V_a + V_c - V_b = 0 \\ V_c = V_d \\ V_a = V_b + V_2 = V_b + 2 \end{cases}$$

$$\begin{cases} V_a = -2V_d \\ 6V_d - 2V_a + 2 = 0 \end{cases} \implies \boxed{\begin{cases} V_a = \dfrac{2}{5} \\ V_b = -\dfrac{8}{5} \\ V_c = -\dfrac{1}{5} \\ V_d = -\dfrac{1}{5} \end{cases}}$$

(h) Draw the superposition circuit for V_3 and write the equation that characterizes this circuit.

Solution:



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$$\begin{cases} \dfrac{V_a - V_1}{R_2} + \dfrac{V_a - V_d}{R_3} + \dfrac{V_b - V_c}{R_4} = 0 \\ \dfrac{V_d}{R_1} + \dfrac{V_d - V_a}{R_3} + \dfrac{V_c - V_b}{R_4} = 0 \\ V_d = V_c + V_3 = V_c + 4 \\ V_a = V_b \\ V_1 = 0 \end{cases} \Longrightarrow \begin{cases} \dfrac{V_a}{2} + \dfrac{V_a - V_d}{4} + \dfrac{V_b - V_c}{4} = 0 \\ V_d + \dfrac{V_d - V_a}{4} + \dfrac{V_c - V_b}{4} = 0 \\ V_d = V_c + 4 V_a = V_b \end{cases}$$

We find that: $\boxed{V_a = -\frac{4}{5}\text{V}, V_b = -\frac{4}{5}\text{V}, V_c = -\frac{18}{5}\text{V}, V_d = \frac{2}{5}\text{V}}$

(i) Find the unknown potentials as you did in in part (d), but now do so using the superposition theorem.

Solution: By the superposition theorem, the actual potential values at nodes V_a through V_d are the linear sums of their corresponding potentials in the simpler circuits. Notice that we kept the ground in the same place for each of the simpler circuits! While the location of the ground does not matter, *keeping it consistent does*. We add up the potentials and find that:

$$\begin{aligned} V_a &= \frac{3}{5} + \frac{2}{5} - \frac{4}{5} = \frac{1}{5} \\ V_b &= \frac{3}{5} - \frac{8}{5} - \frac{4}{5} = -\frac{9}{5} \\ V_c &= \frac{1}{5} - \frac{1}{5} - \frac{18}{5} = -\frac{18}{5} \\ V_d &= \frac{1}{5} - \frac{1}{5} + \frac{2}{5} = \frac{2}{5} \end{aligned}$$

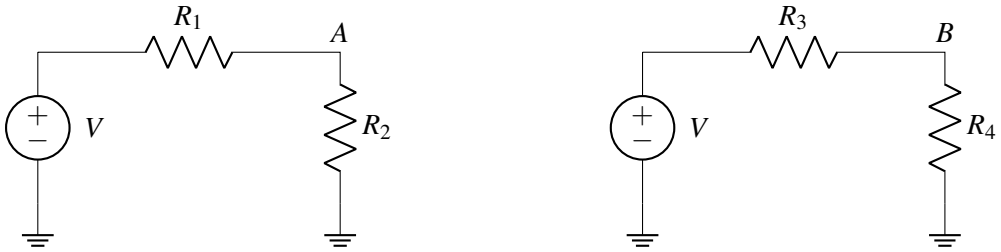
Wow! The same potentials as we found using supernode/nodal analysis!

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2. Modular Circuits

In this question we will explore how to look at circuits in a 'modular' way. That is, we will start to encapsulate circuit components (just like one might encapsulate code into functions) and then see how those components fit together.

(a) Here, you can see two standard voltage dividers. Find the voltage at the points A and B using standard nodal analysis techniques.



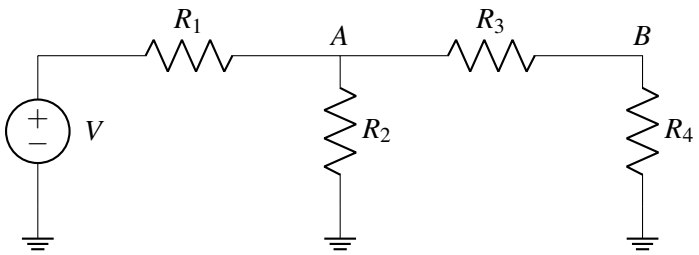
Solution: We can apply KCL at node A. We get:

$$\frac{V_A - 0}{R_2} = \frac{V - V_A}{R_1}$$
$$\implies \frac{R_1}{R_2} V_A = V - V_A$$
$$\implies V_A\left(1 + \frac{R_1}{R_2}\right) = V$$
$$\implies V_A \frac{R_1 + R_2}{R_2} = V$$
$$\implies V_A = V \frac{R_2}{R_1 + R_2}$$

Similarly, for the second circuit, we have

$$V_B = V \frac{R_4}{R_3 + R_4}$$

(b) Now, let us modify this circuit to add a second voltage divider stage starting at A. You can think of this as cascading 2 voltage dividers: one which which has an input of V and an output of V_A, and a second one that has an input of V_A and an output of V_B. Find the voltage at B using nodal analysis.



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Solution: We can apply KCL at node A and at node B. We get the equations:

$$\frac{V-V_A}{R_1}+\frac{0-V_A}{R_2}=\frac{V_A-V_B}{R_3}$$
$$\frac{V_A-V_B}{R_3}=\frac{V_B-0}{R_4}$$

We can simplify these equations to give us

$$\frac{V}{R_1}-\frac{V_A}{R_1}-\frac{V_A}{R_2}=\frac{V_A}{R_3}-\frac{V_B}{R_3}$$
$$\implies \frac{V_B}{R_3}+\frac{V}{R_1}=\frac{V_A}{R_1}+\frac{V_A}{R_2}+\frac{V_A}{R_3}$$
$$\implies \frac{V_B}{R_3}+\frac{V}{R_1}=V_A(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})$$
$$\frac{V_A}{R_3}-\frac{V_B}{R_3}=\frac{V_B}{R_4}$$
$$\implies \frac{V_A}{R_3}=\frac{V_B}{R_3}+\frac{V_B}{R_4}$$
$$\implies V_A=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})$$

Combining these 2 equations, we get:

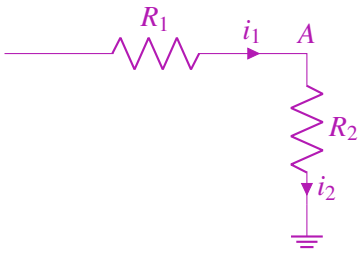
$$\frac{V_B}{R_3}+\frac{V}{R_1}=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})$$
$$\implies \frac{V_B}{R_3}+\frac{V}{R_1}=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})$$
$$\implies \frac{V}{R_1}=V_BR_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})+\frac{V_B}{R_3}=V_B(R_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})+\frac{1}{R_3})$$
$$\implies V_B=\frac{V}{R_1}/(R_3(\frac{1}{R_3}+\frac{1}{R_4})(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})+\frac{1}{R_3})$$

(c) Compare your solwer with what you get when you multiplying the amplification factors of each individual voltage divider. Are they the same or different? Does this surprise you? Why or why not?

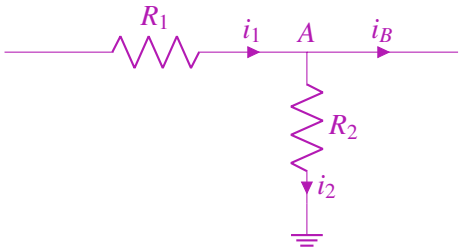
Solution: They are clearly not the same! The next part explains why.

What we see here is called "loading". Ideally, we would like to be able to stack voltage dividers like this in order to cascade their effects. However this does not work because by adding the second circuit, we are changing what the first one does. In particular, the second circuit will draw some current from the first circuit. We can see this difference if we zoom in on the node A. In part (a), we had the following picture:

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And we were able to say that $i_1 = i_2$, because they are the only currents at node A. But after adding the second voltage divider stage, the picture now looks like this:



And so the KCL equation we wrote at node A is no longer valid.

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3. Never fail at resistor equivalence

Solution: Description: derive intuition behind resistor equivalences. Learn about nodes and how to combine resistors in a real problem.

(a) Using what you learned about resistance ($R = \frac{\rho \cdot l}{A}$), explain what would happen if you connect two resistors in series.

Solution: Make sure to draw out resistors as boxes in series and parallel for parts (a) and (b) respectively. The solwer only makes sense once it is drawn out.

Solution: Intuitively, the amount of impedance the current faces by going through the two resistors in series would just be the sum of the impedance faced by the current going through each individual of the resistor. Thus, for n resistors in series,

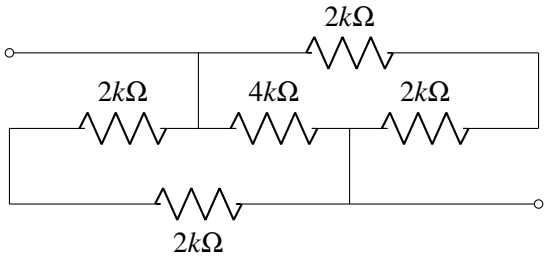
$$R_{eq} = R_1 + R_2 + \cdots + R_n$$

(b) Using the same resistance equation from part (a), explain what would happen if you connect two resistors in parallel.

Solution: Having resistors in parallel is like taking taking two boxes and combining them via their sides. This increases the cross-sectional area in the front. So intuitively, the equivalent resistance of resistors in parallel would be smaller than the resistance of the original resistors. In addition, intuitively, more current would choose to go to the side with less resistance. Thus, for n resistors in parallel,

$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}}$$

(c) Now let's apply some of these ideas to a circuit. For the circuit below, mark all the nodes.



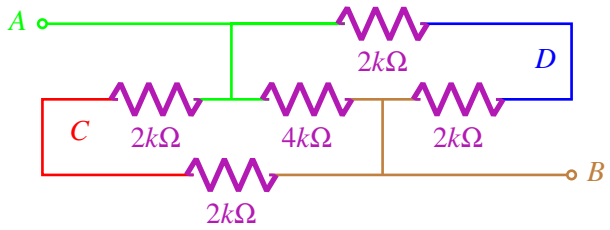
Solution: Mentors: at this point, explain the concept of nodes. Students may be confused as to what constitutes a node and what the properties of a node are.

Here is how I like to teach it: keep a pencil (marker) at a wire, and keep going in all directions till you hit any components. Having multiple colors of markers helps a ton here! Otherwise draw little symbols. Forward slashed lines on one node, backward slashed on another, circles on another and so on.

The property of a node is that the potential on it is the same. The convention is to mark a node with an alphabet, and then call the potential at that point V_a for instance.

Solution:

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(d) Mark which nodes are 2-nodes and multi-nodes. 2 nodes are connected to only 2 components, and multi-nodes are connected to 3 or more components.
Note: *Whenever nodes are marked across which equivalent resistance must be found, those are considered 'components' because something could be connected there.*

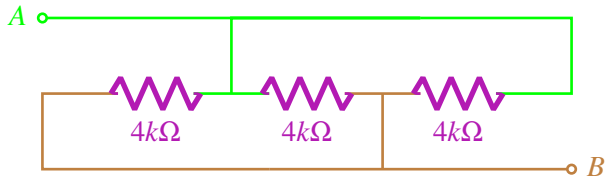
Solution: Multinode and 2-node are not official vocabulary, just aids

Solution: Nodes A and B are multi-nodes. Nodes C and D are 2-nodes.

(e) Resistors that are connected to 2-nodes are considered to be in series. Redraw the circuit, and find the 2 and multi-nodes again after combining the resistors connected to 2 nodes.

Solution:

Nodes C and D disappear and become nodes A and B.



Solution: Mentors please explain how to combine resistors in series. The simple idea is that since they are on a 2-node, you can drag one till you reach the other, and just write $R_1 + R_2$ on that.

(f) Now we should be left with only multi-nodes. So far, we have seen that any resistors connected to 2-nodes are in series. We will see what happens to resistors connected to multi-nodes. Begin by writing out the 2 nodes that each resistor is connected to.

Solution: The resistor on the left is connected to nodes A and B. The resistor on the middle is connected to nodes A and B. The resistor on the right is connected to nodes A and B.

(g) If you have 2 or more resistors that are connected to the same 2 nodes, then they are in parallel. What does this mean for the 3 remaining resistors?

Solution: Since all 3 resistors are connected to nodes A and B, they are all in parallel. The equivalent resistance is $\frac{4}{3}k\Omega$. Quick tip: if you have 2 resistors in parallel, both of whose resistance is R, then the equivalent resistance is $\frac{R}{2}$. Similarly, if you have 3 resistors in parallel, all three of whose resistance is R, then the equivalent resistance is $\frac{R}{3}$, and so on with 4 or more resistors of the same value.

Solution: If students don't realize the tip, show them by deriving it with the formula.

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